

EduGuard Answer Key

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Q1: What is the primary function of photosynthesis in plants?

Answer: To make food from sunlight, water, and air

Q2: Where does the ATP process occur in plant cells?

Answer: In the thylakoid membranes of chloroplasts

Q3: What ions are directly involved in the ATP process of photosynthesis?

Answer: H⁺ (hydrogen ions)

Q4: What role do K⁺ and Na⁺ ions play in plant cells?

Answer: Maintaining the balance of water and salts

Q5: What is the primary energy currency of plant cells?

Answer: ATP (Adenosine Triphosphate)

Q6: What is the function of stomata in plant leaves?

Answer: Gas exchange, allowing carbon dioxide to enter and oxygen to exit

Q7: How do K⁺ ions affect stomatal opening and closing?

Answer: K⁺ ions rush into guard cells, causing them to swell and the stomatal pore to open

Q8: What happens to stomatal function when Na⁺ ions are present in high concentrations?

Answer: Stomatal closure

Q9: Why is the regulation of stomatal aperture important for plants?

Answer: To control gas exchange and water loss, maintaining proper water balance

Q10: Describe the relationship between light energy and ATP production in photosynthesis

Answer: Light energy is absorbed by pigments like chlorophyll, exciting electrons and generating a proton gradient, which drives ATP production

Q11: Explain the role of the electron transport chain in ATP production

Answer: The electron transport chain generates a proton gradient, which drives the production of ATP from ADP and Pi

Q12: What is the outcome of the Calvin cycle, and how is it related to ATP production?

Answer: The Calvin cycle converts CO₂ into glucose, using the ATP produced in the ATP process

Q13: How do plants regulate their water balance, and what role do stomata play in this process?

Answer: Plants regulate their water balance by controlling stomatal aperture, which affects gas exchange and water loss